

RAPID APPLICATION DEVELOPMENT MARKET

GLOBAL FORECAST TO 2023

BY TYPE (LOW-CODE AND NO-CODE), TOOL (MOBILE-BASED, WEB-BASED, DESKTOP-BASED, AND SERVER-BASED), BUSINESS FUNCTION, DEPLOYMENT MODEL, ORGANIZATION SIZE, INDUSTRY VERTICAL, AND REGION

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1 EXECUTIVE SUMMARY

Software engineering has evolved progressively since its inception in 1940s. The methods for building applications have evolved and become better over a period of time. The constant goal to improve technologies and their development practices has enhanced the productivity of business practitioners and quality of applications. Rapid Application Development (RAD) has emerged as a highly interactive application development approach that helps build software applications rapidly.

In today's increasingly competitive marketplace, enterprises have to optimize their performance and processes. The RAD environment helps organizations develop faster and high-quality applications. However, technological developments and user-friendly cloud environments such as Platform-as-a-Service (PaaS), Infrastructure-as-a-Service (IaaS), and Everything-as-a-Service (XaaS) have driven the business mobility on a large scale. This has broadened the scope for the software development domain with the rising Bring your own Device (BYOD) trend, wherein employees can use their personalized devices for the business purpose. Software application development has been in existence since ages, and has significantly enabled the technology industry in the past decade. RAD is a mix of 2 development platforms i.e. the low code development and the no-code development, wherein coders as well as non-coders can create applications based on the inbuilt features with the help of the simple drag-and-drop functionality. Companies save a lot of time and money as non-coders can develop applications with the help of minor coding knowledge. This has provided better benefits to various enterprises in real time. The major factor for the accelerated adoption of RAD tools includes the supportive technology infrastructure, along with increased need for customized software solutions.

Banking, Financial Services, and Insurance (BFSI), manufacturing, and retail and consumer goods are a few verticals that have shown a tremendous adoption of advanced and faster application development platforms to cater to dynamic needs of customers for increasing customer satisfaction. Additionally, organizations of different size have different requirements, where are addressed by RAD tools. A considerable rise has been observed in the mobile-based application development, followed by web-based development, due to the increasing business mobility.

The scope of this report covers the RAD market by type, component, business function, deployment model, organization size, industry vertical, and region. Among components, the tools segment is expected to dominate the market with a higher market share during the forecast period. North America is estimated to hold the highest market share in 2018, owing to the early adoption of advanced software development technologies as well as prominent initiatives taken by the industry players in the form of partnerships with different technology players to offer rapid analytics and data processing solutions. China is one of the developing economies, which has undergone terrific transformation in the areas of manufacturing, telecommunication, and IT.

RAD tool and service vendors have implemented various types of organic as well as inorganic growth strategies, such as new product launches, product upgradations, partnerships and agreements, business expansions, and mergers and acquisitions, to expand their offerings in the market. The major vendors who offer RAD platforms globally are Microsoft Corporation (Microsoft), IBM Corporation (IBM), Oracle Corporation (Oracle), Amazon Web Services, Inc. (AWS), Google Corporation (Google), Mendix, Salesforce.com, inc. (Salesforce), OutSystems, Zoho Corporation Pvt. Ltd. (Zoho Corporation), Appian, Pegasystems Inc. (Pegasystems), ServiceNow, WaveMaker, Inc. (WaveMaker), Ninnox, KiSSFLOW, Radzen Ltd. (Radzen), AmpleLogic.com (AmpleLogic), FileMaker, Inc. (FileMaker), QuickBase, Inc. (QuickBase), AppSheet, LANSA, Kony, Inc. (Kony), SourceCode (K2), MatsSoft, and Oro, Inc. (Oro).

Moreover, enterprises engaged in Artificial Intelligence (AI) and Machine Learning (ML) have partnered with infrastructure providers who provide specialized hardware and software support, and professional expertise to deploy and manage computing workloads. In April 2018, Microsoft partnered with C3 IoT, an AI

and IoT software platform provider, to deliver new technology developments and provide go-to-market strategies that accelerate enterprise AI and IoT application developments. Combining the market reach of Microsoft Azure and the capability of C3 IoT's AI platform, Microsoft plans to accelerate the adoption of AI, IoT, and elastic cloud computing globally. Later in November 2018, Zoho Corporation launched two products: Zoho PageSense, a website optimization software that helps marketers expand conversion rates through visualizing, testing, and measuring behavior patterns, and Zoho Flow, a drag-and-drop integration platform that assists users in creating flexible workflows between cloud applications.

The RAD market is expected to grow from USD XX million in 2018 to USD XX million by 2023, at a CAGR of XX% during the forecast period.

TABLE 1 RAPID APPLICATION DEVELOPMENT MARKET SIZE AND GROWTH RATE, 2016–2023 (USD MILLION, Y-O-Y %)

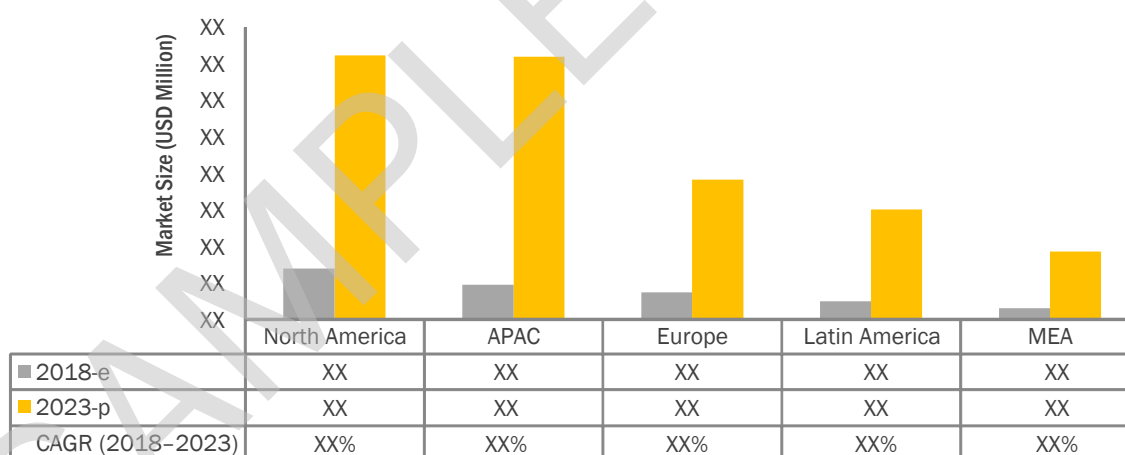
RAD Market	2016	2017	2018-e	2023-p	CAGR (2018–2023)
Market Size	XX	XX	XX	XX	XX%
Y-o-Y	XX	XX	XX	XX	XX%

e: estimated; p: projected

Note: The Y-o-Y for 2023 is calculated on the basis of the values projected for 2022

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 1 RAPID APPLICATION DEVELOPMENT MARKET TO WITNESS HIGH GROWTH DURING THE FORECAST PERIOD



e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

2 MARKET OVERVIEW AND INDUSTRY TRENDS

2.1 INTRODUCTION

The demand for RAD is expected to rise over the next 5 years, owing to various factors, such as the increase in enterprise mobility, and this has led to the increase in the number of smartphones used for businesses. The other factor is the need for rapid customization, especially when there is massive data generation across the BFSI, retail, and media and entertainment industries, and the availability of open source solutions and growing application areas. The high demand for solutions enabling high-performance computing in various industries creates ample opportunities for the adoption of RAD tools and services, and the RAD vendors can monetize such opportunities. In addition, the lack of sufficient technical expertise and domain knowledge hinder the growth of the market. Moreover, data security and limited capabilities of the available RAD tools in performing crucial operations in analytics use-cases prove to be challenging to the growth of the market. This section discusses the drivers, restraints, opportunities, and challenges involved in the RAD market and describes the impact of these parameters on the market.

2.2 MARKET DYNAMICS

The following section highlights the drivers, restraints, opportunities, and challenges in the RAD market:

FIGURE 2 DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES: RAPID APPLICATION DEVELOPMENT MARKET

 DRIVERS	▪ Increasing need for rapid customization and scalability ▪ Enterprise mobility to drive the adoption of smartphones in businesses ▪ Elimination of gaps in the required IT skills
 RESTRAINTS	▪ Dependency on vendor-supplied customization
 OPPORTUNITIES	▪ Growing digital transformation in the IT industry
 CHALLENGES	▪ Legacy software in need of upgradation

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

3 RAPID APPLICATION DEVELOPMENT MARKET, BY TYPE

3.1 LOW-CODE DEVELOPMENT PLATFORMS

3.1.1 ORGANIZATIONAL NEED TO DEVELOP CUSTOMER-CENTRIC APPLICATIONS

Many technology vendors are increasingly adopting low-code development platforms. The platform provides a simple interface that enables the seamless integration of various components with third-party APIs for testing. Additionally, the platform enables citizen developers to design and create web and mobile applications using minimum hand-coding. Enterprises are rapidly deploying low-code development platforms to develop customer-centric applications quickly. Moreover, the platform provides various benefits, such as low initial setup costs and easy development, which are expected to be increasing the popularity of the platform among enterprise customers.

TABLE 2 LOW-CODE DEVELOPMENT PLATFORMS: RAPID APPLICATION DEVELOPMENT MARKET SIZE, BY REGION, 2016–2023 (USD MILLION)

Region	2016	2017	2018-e	2023-p	CAGR (2018–2023)
North America	XX	XX	XX	XX	XX%
Europe	XX	XX	XX	XX	XX%
APAC	XX	XX	XX	XX	XX%
MEA	XX	XX	XX	XX	XX%
Latin America	XX	XX	XX	XX	XX%
Total	XX	XX	XX	XX	XX%

e: estimated, p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the low-code development platforms segment by region. The total market is expected to grow from USD XX million in 2018 to USD XX million by 2023, at a CAGR of XX% during the forecast period. APAC is expected to witness considerable growth in the market and is projected to be the second highest revenue-generating region in 2018, after North America. Also, APAC is expected to reach a market size of USD XX million by 2023, growing at a CAGR of XX%.

4 RAPID APPLICATION DEVELOPMENT MARKET, BY COMPONENT

4.1 TOOLS

The RAD methodology has simplified the application development process by helping developers design applications without using hard coding or technical expertise. Programming with minimum code or no coding has revolutionized the application development life cycle and expanded its reach across various platforms, including mobiles, web, desktops, and servers. These platforms are adopted by citizen developers and technically trained developers to create customized applications. RAD tools and services offer features, such as the drag-and-drop functionality, cross-platform development capability, visual development approach, quick deployment facilities, low costs, and no additional infrastructure fee. These capabilities that the RAD tools offer help in designing high-quality applications, saving time, and eliminating additional associated costs.

TABLE 3 TOOLS: RAPID APPLICATION DEVELOPMENT MARKET SIZE, BY TYPE, 2016–2023 (USD MILLION)

Type	2016	2017	2018-e	2023-p	CAGR (2018–2023)
Mobile-based	XX	XX	XX	XX	XX%
Web-based	XX	XX	XX	XX	XX%
Desktop-based	XX	XX	XX	XX	XX%
Server-based	XX	XX	XX	XX	XX%
Total	XX	XX	XX	XX	XX%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the tools segment by type. The mobile-based segment is expected to grow from USD XX million in 2018 to USD XX million by 2023, at a CAGR of XX% during the forecast period. The web-based segment is expected to hold the largest market share of XX in 2023 and grow at the second highest CAGR of XX%.

4.1.1 MOBILE-BASED

4.1.1.1 The increase in the number of mobile device users calls for more mobile-based applications

Mobile-based application development tools help users design application using smartphones. Business owners are increasingly adopting RAD tools to simplify the application development process and cater to their customers' needs using mobile-based applications. Mobile applications require specific software that are designed to run on mobile devices, such as smartphones, tablets, and smartwatches. Due to the increasing number of mobile device users, mobile applications are also gaining traction. Organizations are trying to expand their presence by offering mobile-based services to their customers. Mobile application developers are designing customized mobile applications for various OS, such as iOS, Android, and Windows. The upgradation of mobile applications is different from that of web-based applications, as every application needs to be upgraded on individual mobile devices. The increasing popularity of mobile applications is attracting developers to offer applications for various OS. Such factors are expected to encourage the deployment of RAD tools across the globe.

5 RAPID APPLICATION DEVELOPMENT MARKET, BY DEPLOYMENT MODEL

5.1 CLOUD

5.1.1 SCALABILITY, FLEXIBILITY, AND RELIABILITY BENEFITS

The cloud deployment model is expected to be adopted more for deploying RAD platforms. SMEs, in particular, implement RAD platforms using the cloud deployment model, as it enables them to focus on their core competencies rather than investing more capital on their IT infrastructure. Using the cloud deployment model for deploying the suitable RAD platform, organizations can avoid costs related to hardware, software, storage, and technical staff. In fact, the cloud-based deployment model helps in the rapid development of business applications, which, in turn, is beneficial for organizations that have strict budgets for investments in IT infrastructure. Moreover, cloud-based RAD platforms are easy to maintain and upgrade, thereby leading to the growth of this deployment model.

TABLE 4 CLOUD: RAPID APPLICATION DEVELOPMENT MARKET SIZE, BY REGION, 2016–2023 (USD MILLION)

Region	2016	2017	2018-e	2023-p	CAGR (2018–2023)
North America	XX	XX	XX	XX	XX%
Europe	XX	XX	XX	XX	XX%
APAC	XX	XX	XX	XX	XX%
MEA	XX	XX	XX	XX	XX%
Latin America	XX	XX	XX	XX	XX%
Total	XX	XX	XX	XX	XX%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the cloud deployment model by region. MEA is expected to be the second fastest region in terms of growth and estimated to grow from USD XX million in 2018 to reach USD XX million by 2023, at a CAGR of XX% during the forecast period. APAC is projected to grow at the highest CAGR of XX% during the forecast period.

5.2 ON-PREMISES

5.2.1 NEED FOR HIGHLY SECURE OPERATING ENVIRONMENT

The on-premises deployment model is the traditional approach to implementing RAD platforms. It provides organizations with full control over the platform, applications, systems, and data, and these can be handled and managed by the organization's own IT staff. An organization that handles customers' credentials would prefer the on-premises deployment model, as the systems can be controlled by the organization's staff themselves. The government and BFSI verticals prefer the on-premises deployment model, as they cannot take the risk of exposing the confidential data of the nation and losing sensitive data. Moreover, they prefer confidentiality for financial records, accounting information, and money transaction information.

6 RAPID APPLICATION DEVELOPMENT MARKET, BY ORGANIZATION SIZE

6.1 SMALL AND MEDIUM-SIZED ENTERPRISES

6.1.1 THE NEED FOR COST-EFFECTIVE SOLUTIONS TO MEET BUDGETS AND LEAD TO BUSINESS GROWTH

According to the definition given by the European Union (EU), the SMEs comprise less than 250 employees, whereas the US defines SMEs as organizations with less than 500 employees; the number varies from region to region. According to this research study, the SMEs have an employee size of 1–1,000. The SMEs are mainly adopting RAD platforms to develop business applications and improve their productivity rapidly. One of the prime motives of the SMEs is to develop customer-centric applications at lower costs. Moreover, the SMEs are looking for cloud-based RAD platforms to save money, time, and resources. Another reason for the SMEs to prefer cloud-based solutions is their financial constraints. Such features are compelling the SMEs to adopt cloud-based RAD platforms, which is expected to be driving the overall growth of the RAD market.

TABLE 5 SMALL AND MEDIUM-SIZED ENTERPRISES: RAPID APPLICATION DEVELOPMENT MARKET SIZE, BY REGION, 2016–2023 (USD MILLION)

Region	2016	2017	2018-e	2023-p	CAGR (2018–2023)
North America	XX	XX	XX	XX	XX%
Europe	XX	XX	XX	XX	XX%
APAC	XX	XX	XX	XX	XX%
MEA	XX	XX	XX	XX	XX%
Latin America	XX	XX	XX	XX	XX%
Total	XX	XX	XX	XX	XX%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the SMEs segment by region. North America is expected to grow from USD XX million in 2018 to USD XX million by 2023, at a CAGR of XX% during the forecast period. The APAC region is expected to expand at the highest CAGR XX% during the forecast period.

7 RAPID APPLICATION DEVELOPMENT MARKET, BY INDUSTRY VERTICAL

7.1 BANKING, FINANCIAL SERVICES, AND INSURANCE

7.1.1 THE NEED TO DELIVER ENRICHED CUSTOMER EXPERIENCE USING NEW FINANCIAL PRODUCTS AND SERVICES

This industry vertical comprises commercial banks, insurance companies, mutual funds, cooperatives, non-banking financial companies, pension funds, and other smaller financial institutions. It is an early adopter of cutting-edge applications, as it needs to engage a large customer base that holds highly sensitive financial data. Therefore, this industry vertical faces different challenges related to stringent regulatory compliances and security requirements in terms of providing superior service to its customers. In addition, the industry vertical is always on the lookout for efficient and secure applications and services that can help deliver an enriched experience to customers and protect its employees, customers, assets, offices, branches, and operations. Such factors are expected to drive the overall RAD market size. Moreover, this industry vertical frequently keeps introducing new and improved financial products and services to enhance its business operations, which makes it attractive for the RAD platform providers to offer advanced solutions to the industry vertical. With facilities such as smart banking, internet banking, and mobile banking, the BFSI industry vertical has to prioritize on robust and rapid application designs and their development.

TABLE 6 BANKING, FINANCIAL SERVICES, AND INSURANCE: RAPID APPLICATION DEVELOPMENT MARKET SIZE, BY REGION, 2016–2023 (USD MILLION)

Region	2016	2017	2018-e	2023-p	CAGR (2018–2023)
North America	XX	XX	XX	XX	XX%
Europe	XX	XX	XX	XX	XX%
APAC	XX	XX	XX	XX	XX%
MEA	XX	XX	XX	XX	XX%
Latin America	XX	XX	XX	XX	XX%
Total	XX	XX	XX	XX	XX%

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the BFSI industry vertical by region. North America is expected to grow from USD XX million in 2018 to USD XX million by 2023, at a CAGR of XX% during the forecast period. APAC is projected to grow at the highest CAGR of XX% during the forecast period.

8 RAPID APPLICATION DEVELOPMENT MARKET, BY REGION

8.1 INTRODUCTION

North America has the highest adoption rate of new technologies compared to the other regions and is expected to hold the largest market in the RAD market, due to the presence of large enterprises, technical expertise, application modernization service providers, and cloud service providers, and the trend of the companies in the region, which is to adopt the latest technologies. APAC, MEA, and Latin America are showing an exponential growth rate in the adoption of RAD platforms, as many large enterprises are observing a huge potential to expand their market to these regions. SMEs in these regions have also realized the benefits of cloud computing and are migrating their applications to cloud to drive business results and improve their efficiency and productivity. This section provides region-wise industry patterns, market potential, and forecasts based on types, components, deployment models, business functions, organization size, and industry verticals.

TABLE 7 RAPID APPLICATION DEVELOPMENT MARKET SIZE, BY REGION, 2016–2023 (USD MILLION)

Region	2016	2017	2018-e	2023-p	CAGR (2018–2023)
North America	XX	XX	XX	XX	XX%
Europe	XX	XX	XX	XX	XX%
APAC	XX	XX	XX	XX	XX%
MEA	XX	XX	XX	XX	XX%
Latin America	XX	XX	XX	XX	XX%
Total	XX	XX	XX	XX	XX%

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Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

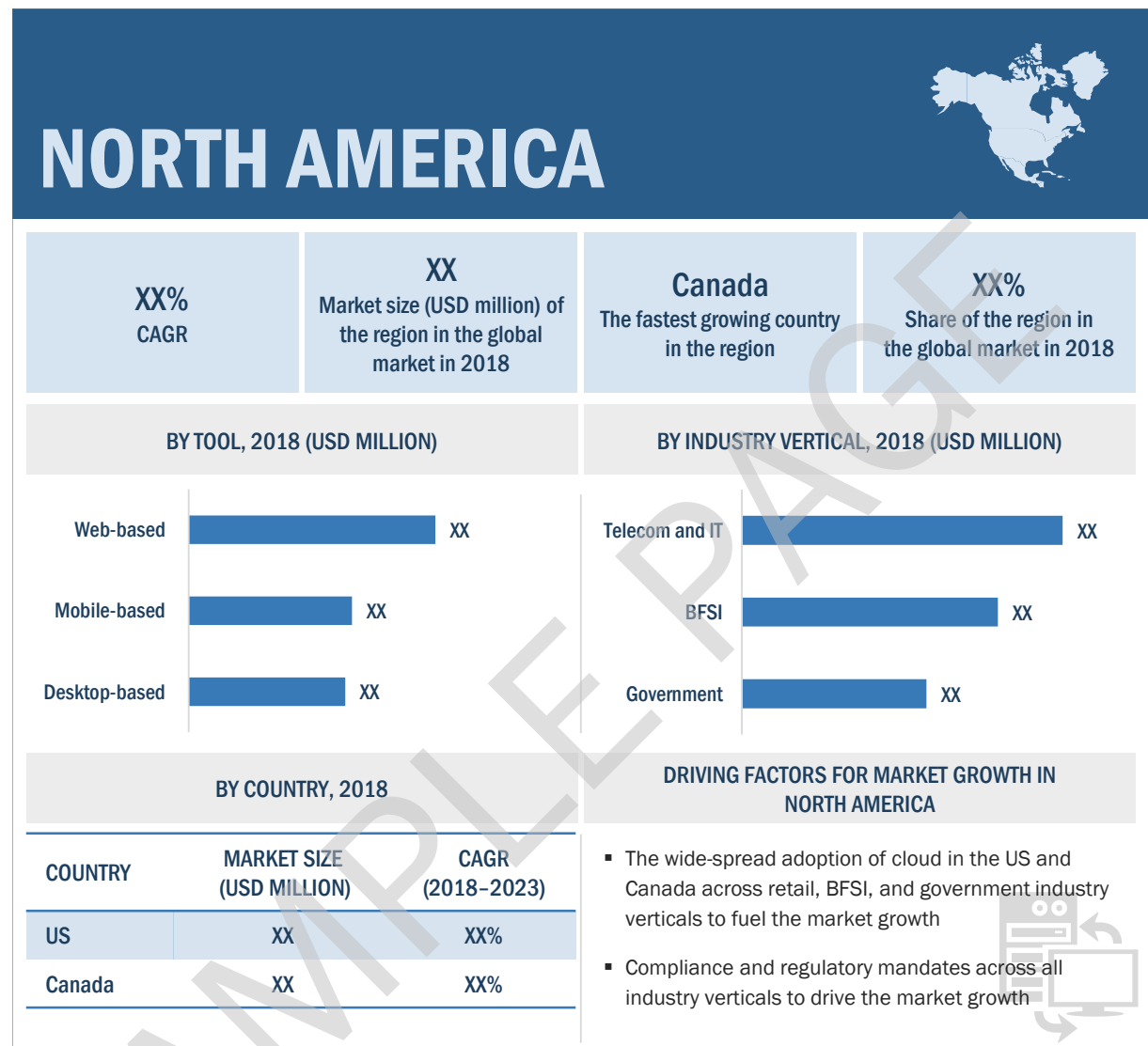
The table above highlights the RAD market size by region. North America is estimated to hold the largest market size of USD XX million in 2018, which is projected to reach USD XX million by 2023, at the CAGR of XX%. Whereas, continuous developments and the adoption of various technologies across all the major industry verticals are expected to drive the growth of the market in APAC. The growing adoption of cloud infrastructure, especially among the SMEs, is expected to increase the demand for RAD platforms in the APAC region. APAC is expected to reach a market size of USD XX million by 2023, growing at a CAGR of XX% during the forecast period.

8.2 NORTH AMERICA

North America is estimated to contribute the largest share to the global RAD market. Due to the emergence of trending technologies, such as IoT, AI, ML, cloud computing, big data, DevOps, and enterprise mobility, users and enterprises have been urged to adopt much more sophisticated and reliable application platform software and the associated services. This fact is expected to contribute to the overall growth of the RAD market in North America. Additionally, strong healthcare and BFSI compliances, high penetration of advanced cloud technologies, and better citizen services are helping the region grow in terms of adopting the emerging technologies. The North American region comprises the US and Canada, which are also witnessing a phenomenal adoption of RAD platforms. These countries have a major dominance with sustainable and well-established economies, empowering them to strongly invest in R&D activities, thereby contributing to the development of new technologies. Moreover, the governments in this region have a

strong vision and budget allocations to modernize their legacy systems, which is helping organizations grow rapidly.

FIGURE 3 NORTH AMERICA: MARKET SNAPSHOT



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

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